Phpunit Test Documentation

SaltOS 4.1 r2774

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1 Phpunit Libraries

1.1 Autoload file for the unit tests

```
lib/autoload.php
```

This file contains the code that initialize the unit tests

1.1.1 Importing namespaces

```
use PHPUnit\Framework\Assert;
```

1.1.2 Main autoloader code

This code emmulates the index.php by loading all autoload files excep the zindex.php and initialize the timer

1.2 Unit test helper functions

```
lib/utestlib.php
```

This file contains the functions used by the web and cli unit tests to get the coverage, too contains the pcov and mime helpers.

1.2.1 Importing namespaces

```
use SebastianBergmann\CodeCoverage\Data\RawCodeCoverageData;
```

1.2.2 Define the variable used to add this directory to the list of allowed paths,

```
define('K_ALLOWED_PATHS', [dirname(dirname(getcwd()))]);
```

1.2.3 Test PCOV start helper

```
function test_pcov_start(): void
```

This function puts the flag needed to enable the pcov feature, to do it, the function creates a void file and puts permissions to allow that other procs can write inside of it

1.2.4 Test PCOV stop helper

```
function test_pcov_stop($index): void
```

This function gets the contents of the coverage file and removes it, too, it does all needed things to append the collected coverage to the current unit test coverage instance

- @index => index used to get the backtrace, it depends from where you are calling this function, generally is 2 but in some cases you need to use another value like 1.

1.2.5 Test WEB helper

```
function test_web_helper($rest, $data, $token, $lang)
```

This function performs the action defined by the rest verb sendind the data if it is provided and using the token for authentication actions.

As you can see in the code, the function detects if data is provided and send the request using GET or POST, in addition, an application/json content-type header is send when POST is used.

The token is sent using the TOKEN header to be used in the authentication process.

- @rest => The rest request, like update/customers/3
- @data => The data used as json in the SaltOS app
- @token => The token used if authentication is required

1.2.6 Test CLI helper

```
function test_cli_helper($rest, $data, $token, $lang, $user)
```

This function allow to execute SaltOS using the CLI SAPI, to do it, the function detects if data is provided, and executes the command and getting the output of the execution. If data exists, then the contents are stored in a file and passed the contents of the file to the stdin of the php process to emmulate the input channel used by the apache server.

As an example, this functions tries to execute the command using the follow formula:

1) php index.php $rest

2) cat /tmp/input | php index.php $rest

In addition, the token field is used to define the TOKEN environment variable that is used by SaltOS as variable to emmulate the TOKEN used by the apache for authenticate the SaltOS app.

- @rest => The rest request, like update/customers/3
- @data => The data used as json in the SaltOS app
- @token => The token used if authentication is required

1.2.7 Get Mime helper

```
function get_mime($buffer): string
```

This function returns the content type of the contents of the buffer

- @buffer => the contents that you want to check

1.2.8 External tests

```
function test_external_exec($test_file, $error_file, $error_words): void
```

This function tries to execute external php that triggers special conditions suck as errors, that are not allowed from the phpunit tests

- @test_file => the file that must be executed
- @error_file => the file that must contains the error_words
- @error_words => the words that must exists in the error_file

2 Phpunit Code

2.1 Test about

```
test_about.php
```

This test performs some tests to validate the correctness of the about feature

2.1.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.1.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.1.3 Main class of this unit test

```
final class test_about extends TestCase
```

2.1.4 about test

```
public function test_about(): void
```

This test performs some tests to validate the correctness of the about feature

2.2 Test actions

```
test_actions.php
```

This test performs some tests to validate the correctness of the actions functions

2.2.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.2.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.2.3 Main class of this unit test

```
final class test_actions extends TestCase
```

2.2.4 actions test

```
public function test_actions(): void
```

This test performs some tests to validate the correctness of the actions functions

2.3 Test addlog

```
test_add.php
```

This test performs some tests to validate the correctness of the addlog functions

2.3.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.3.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.3.3 Main class of this unit test

```
final class test_add extends TestCase
```

2.3.4 addlog test

```
public function test_addlog(): void
```

This test performs some tests to validate the correctness of the addlog functions

2.3.5 adderror test

```
public function test_adderror(): void
```

This test performs some tests to validate the correctness of the adderror functions

2.4 Test apps

```
test_apps.php
```

This test performs some tests to validate the correctness of the apps functions

2.4.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.4.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.4.3 Main class of this unit test

```
final class test_apps extends TestCase
```

2.4.4 apps test

```
public function test_apps(): void
```

This test performs some tests to validate the correctness of the apps functions

2.4.5 app test

```
public function test_app(): void
```

This test performs some tests to validate the correctness of the app functions

2.4.6 list test

```
public function test_list(): void
```

This test performs some tests to validate the correctness of the list functions

2.5 Test array

```
test_array.php
```

This test performs some tests to validate the correctness of the array functions

2.5.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.5.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.5.3 Main class of this unit test

```
final class test_array extends TestCase
```

2.5.4 array test

```
public function test_array(): void
```

This test performs some tests to validate the correctness of the array functions

2.6 Test ascii

```
test_ascii.php
```

This test performs some tests to validate the correctness of the ascii functions

2.6.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.6.2 Loading helper function

```
require_once 'php/lib/ascii.php';
```

This file contains the needed function used by the unit tests

2.6.3 Main class of this unit test

```
final class test_ascii extends TestCase
```

2.6.4 ascii test

```
public function test_ascii(): void
```

This test performs some tests to validate the correctness of the ascii functions

2.7 Test authrenew

```
test_authrenew.php
```

This test performs some tests to validate the correctness of the authrenew functions

2.7.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.7.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.7.3 Main class of this unit test

```
final class test_authrenew extends TestCase
```

2.7.4 authrenew test

```
public function test_authrenew(): void
```

This test performs some tests to validate the correctness of the authrenew functions

2.8 Test authupdate

```
test_authupdate.php
```

This test performs some tests to validate the correctness of the authupdate functions

2.8.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.8.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.8.3 Main class of this unit test

```
final class test_authupdate extends TestCase
```

2.8.4 authupdate test

```
public function test_authupdate(): void
```

This test performs some tests to validate the correctness of the authupdate functions

2.9 Test cli customers

```
test_cli_customers.php
```

This test performs all actions of the customers app suck as: create, insert, list, view, edit, update and delete, using the cli sapi interface

2.9.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.9.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.9.3 Main class of this unit test

```
final class test_cli_customers extends TestCase
```

2.9.4 Authtoken

```
public function test_authtoken(): array
```

This function execute the authtoken rest request, and must to get the json with the valid token to continue in the nexts unit tests

2.9.5 Create

```
public function test_create(array $json): array
```

This function execute the creates rest request, and must to get the json with the layout without data

2.9.6 Insert

```
public function test_insert(array $json): array
```

This function execute the insert rest request, to do it send the json with the data that they want to insert and must to get the json with the status and the create_id.

2.9.7 List

```
public function test_list(array $json): array
```

This function execute the list rest request, to do it send the json with the search that they want to use in the list filter and receives the json with the data used to populate the table.

2.9.8 View

```
public function test_view(array $json): array
```

This function execute the view rest request, intended to retrieve the detail of the app with the layout needed to render it.

2.9.9 Edit

```
public function test_edit(array $json): array
```

This function execute the view rest request, intended to retrieve the detail of the app with the layout needed to render it.

2.9.10 Upgrade

```
public function test_update(array $json): array
```

This function execute the update rest request, to do it send the json with the data that they want to update and must to get the json with the status and the updated_id.

2.9.11 Delete

```
public function test_delete(array $json): void
```

This function execute the delete rest request, they must to get the json with the status and the deleted_id.

2.10 Test cli invoices

```
test_cli_invoices.php
```

This test performs all actions of the invoices app suck as: create, insert, list, view, edit, update and delete, using the cli sapi interface

2.10.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.10.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.10.3 Main class of this unit test

```
final class test_cli_invoices extends TestCase
```

2.10.4 Authtoken

```
public function test_authtoken(): array
```

This function execute the authtoken rest request, and must to get the json with the valid token to continue in the nexts unit tests

2.10.5 Create

```
public function test_create(array $json): array
```

This function execute the creates rest request, and must to get the json with the layout without data

2.10.6 Insert

```
public function test_insert(array $json): array
```

This function execute the insert rest request, to do it send the json with the data that they want to insert and must to get the json with the status and the create_id.

2.10.7 List

```
public function test_list(array $json): array
```

This function execute the list rest request, to do it send the json with the search that they want to use in the list filter and receives the json with the data used to populate the table.

2.10.8 View

```
public function test_view(array $json): array
```

This function execute the view rest request, intended to retrieve the detail of the app with the layout needed to render it.

2.10.9 Edit

```
public function test_edit(array $json): array
```

This function execute the view rest request, intended to retrieve the detail of the app with the layout needed to render it.

2.10.10 Upgrade

```
public function test_update(array $json): array
```

This function execute the update rest request, to do it send the json with the data that they want to update and must to get the json with the status and the updated_id.

2.10.11 Delete

```
public function test_delete(array $json): void
```

This function execute the delete rest request, they must to get the json with the status and the deleted_id.

2.11 Test cli tokens (first part)

```
test_cli_tokens.php
```

This test performs some part of the actions related with the tokens suck as authtoken and checktoken, using the cli sapi interface

2.11.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.11.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.11.3 Main class of this unit test

```
final class test_cli_tokens extends TestCase
```

2.11.4 Authtoken

```
public function test_authtoken(): array
```

This function execute the authtoken rest request, and must to get the json with the valid token to continue in the nexts unit tests

2.11.5 Checktoken

```
public function test_checktoken(array $json): array
```

This function execute the checktoken rest request, and must to get the json with the ok about the valid token that you are trying to check

2.11.6 Deauthtoken

```
public function test_deauthtoken(array $json): array
```

This function execute the deauthtoken rest request, and must to get the json with the ok about the valid token that you are deauthenticate

2.11.7 Checktoken ko

```
public function test_checktoken_ko(array $json): void
```

This function execute the checktoken rest request, and must to get the json with the ko about the invalid token that you are trying to check

2.12 Test color

```
test_color.php
```

This test performs some tests to validate the correctness of the color feature

2.12.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.12.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.12.3 Main class of this unit test

```
final class test_color extends TestCase
```

2.12.4 color test

```
public function test_color(): void
```

This test performs some tests to validate the correctness of the color feature

2.13 Test common

```
test_common.php
```

This test performs some tests to validate the correctness of the common functions

2.13.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.13.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.13.3 Main class of this unit test

```
final class test_common extends TestCase
```

2.13.4 common test

```
public function test_common(): void
```

This test performs some tests to validate the correctness of the common functions

2.13.5 make_app_file test

```
public function test_make_app_file(): void
```

This test performs some tests to validate the correctness of the make_app_file function, used to convert a quick app yaml spec into its equivalent xml file

2.14 Test config

```
test_config.php
```

This test performs some tests to validate the correctness of the config functions

2.14.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.14.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.14.3 Main class of this unit test

```
final class test_config extends TestCase
```

2.14.4 config test

```
public function test_config(): void
```

This test performs some tests to validate the correctness of the config functions

2.15 Test control

```
test_control.php
```

This test performs some tests to validate the correctness of the control functions

2.15.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.15.2 Loading helper function

```
require_once 'php/lib/control.php';
```

This file contains the needed function used by the unit tests

2.15.3 Main class of this unit test

```
final class test_control extends TestCase
```

2.15.4 control test

```
public function test_control(): void
```

This test performs some tests to validate the correctness of the control functions

2.15.5 log test

```
public function test_log(): void
```

This test performs some tests to validate the correctness of the log functions

2.15.6 version test

```
public function test_version(): void
```

This test performs some tests to validate the correctness of the version functions

2.15.7 indexing test

```
public function test_indexing(): void
```

This test performs some tests to validate the correctness of the indexing functions

2.15.8 integrity test

```
public function test_integrity(): void
```

This test performs some tests to validate the correctness of the integrity function, that periodically scans every app table looking for drift against its control table and repairs it via make_control()

2.16 Test crm

```
test_crm.php
```

This test performs some tests to validate the correctness of the quotes app functions (unmake_matrix_data and set_quote), using the web interface since these functions are loaded on demand by the XML router and can't be required directly (their name clashes with other apps that define a function with the same name, such as apps/users/php/matrix.php)

2.16.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.16.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.16.3 Main class of this unit test

```
final class test_crm extends TestCase
```

2.16.4 Authtoken

```
public function test_authtoken(): array
```

This function execute the authtoken rest request, and must to get the json with the valid token to continue in the nexts unit tests

2.16.5 quotes test

```
public function test_quotes(array $json): void
```

This test performs some tests to validate the correctness of the unmake_matrix_data and set_quote functions, exercised through the insert and update rest actions of the quotes app

2.16.6 quotes matrix edge cases test

```
public function test_quotes_matrix_edge_cases(array $json): void
```

This test performs some tests to validate the correctness of the unmake_matrix_data and set_quote functions when they receive edge case inputs: an empty payload, a fully blank new row (must be dropped silently), a blank existing row (must become a deletion marker) and a leftover row beyond the submitted matrix (must also become a deletion marker)

2.16.7 quotes withholding test

```
public function test_quotes_withholding(array $json): void
```

This test performs some tests to validate the correctness of the unmake_matrix_data function when the lines are using taxes and withholdings, the withholdings must be stored in the lines, in the taxes breakdown using the withholding type and in the totals

2.16.8 quotes numbering test

```
public function test_quotes_numbering(array $json): void
```

This test performs some tests to validate the correctness of the numbering of the quotes, the next code must be computed using the numeric value of the codes of the current year, and not their text

2.17 Test cron

```
test_cron.php
```

This test performs some tests to validate the correctness of the cron functions

2.17.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.17.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.17.3 Main class of this unit test

```
final class test_cron extends TestCase
```

2.17.4 Wait cron

```
private function wait_cron(): void
```

This function waits 10 seconds until all cron processes are running

2.17.5 cron

```
public function test_cron(): void
```

This test performs some tests to validate the correctness of the cron functions

2.18 Test customers

```
test_customers.php
```

This test performs some tests to validate the correctness of the customers functions

2.18.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.18.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.18.3 Main class of this unit test

```
final class test_customers extends TestCase
```

2.18.4 customers test

```
public function test_customers(): void
```

This test performs some tests to validate the correctness of the customers functions

2.19 Test dashboard

```
test_dashboard.php
```

This test performs some tests to validate the correctness of the dashboard functions

2.19.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.19.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.19.3 Main class of this unit test

```
final class test_dashboard extends TestCase
```

2.19.4 Authtoken

```
public function test_authtoken(): array
```

This function execute the authtoken rest request, and must to get the json with the valid token to continue in the nexts unit tests

2.19.5 dashboard test

```
public function test_dashboard(array $json): void
```

This test performs some tests to validate the correctness of the dashboard functions

2.19.6 Collect helper

```
private function collect(array $items, string $name): array
```

xml2array()/set_array() store repeated elements as "name", "name#1", "name#2", … instead of a plain list, this helper flattens them back into a plain list of their #attr arrays, to ease the assertions above

2.20 Test database drivers

```
test_database.php
```

2.20.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.20.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.20.3 Main class of this unit test

```
final class test_database extends TestCase
```

2.20.4 Helper

```
private function test_helper($obj): void
```

This function executes the follow queries and checks the correctness of the driver by comparing the results with the expected results.

The tests that performs are the follow:

- SELECT GROUP_CONCAT(a) test FROM (SELECT 1 a UNION SELECT 2 a UNION SELECT 3 a) a;
- SELECT REPLACE('abc', 'b', 'c') test
- SELECT LPAD('123', '5', '0') test
- SELECT CONCAT('a', 'b', 'c') test
- SELECT CONCAT_WS(',','a','b','c',null,true,false) test
- SELECT UNIX_TIMESTAMP('2024-02-01 12:34:56') test
- SELECT FROM_UNIXTIME(1706787296) test
- SELECT YEAR('2024-02-01 12:34:56') test
- SELECT MONTH('2024-02-01 12:34:56') test
- SELECT WEEK('2024-02-01 12:34:56', 1) test
- SELECT TRUNCATE(1.2345, 2) test
- SELECT DAY('2024-02-01 12:34:56') test
- SELECT DAYOFYEAR('2024-02-01 12:34:56') test
- SELECT DAYOFWEEK('2024-02-01 12:34:56') test

- SELECT HOUR('2024-02-01 12:34:56') test
- SELECT MINUTE('2024-02-01 12:34:56') test
- SELECT SECOND('2024-02-01 12:34:56') test
- SELECT MD5('fortuna') test
- SELECT REPEAT('abc',3) test
- SELECT FIND_IN_SET(3,'1,2,3,4,5') test
- SELECT FIND_IN_SET(6,'1,2,3,4,5') test
- SELECT FIND_IN_SET(3,'12345') test
- SELECT IF(true, 'ok', 'ko') test
- SELECT IF(false, 'ok', 'ko') test
- SELECT IF(null, 'ok', 'ko') test
- SELECT POW(2, 8) test
- SELECT DATE_FORMAT('2024-02-01 12:34:56', '%Y-%m-%d %H:%i:%s') test
- SELECT NOW() test

2.20.5 Last Insert Id helper

```
private function last_insert_id_helper($obj): void
```

This function is intended to check the correctness of the last_insert_id feature implemented in each database driver.

2.20.6 PDO MySQL driver

```
public function test_pdo_mysql(): void
```

This function checks the correctness of the sqlite3 driver by creating a database connection, sendint queries validating the expected results and closing the connection.

2.20.7 MySQL improved driver

```
public function test_mysqli(): void
```

This function checks the correctness of the sqlite3 driver by creating a database connection, sendint queries validating the expected results and closing the connection.

2.20.8 PDO SQLite driver

```
public function test_pdo_sqlite(): void
```

This function checks the correctness of the sqlite3 driver by creating a database connection, sendint queries validating the expected results and closing the connection.

2.20.9 SQLite3 driver

```
public function test_sqlite3(): void
```

This function checks the correctness of the sqlite3 driver by creating a database connection, sendint queries validating the expected results and closing the connection.

2.20.10 PDO mssql driver

```
public function test_pdo_mssql(): void
```

This function checks the correctness of the sqlserver driver by creating a database connection, sendint queries validating the expected results and closing the connection.

2.20.11 PDO pgsql driver

```
public function test_pdo_pgsql(): void
```

This function checks the correctness of the sqlserver driver by creating a database connection, sendint queries validating the expected results and closing the connection.

2.20.12 Not null helper

```
private function not_null_helper($obj, $extra, $asis): void
```

This function validates how the rows of a table that allows nulls are copied to a table that not allows nulls and have a default value, as the dbschema does when the structure of a table changes, the copy of the field as is only works in some engines, and by this reason the dbschema uses the ifnull to convert the nulls to the default value.

- @obj => the database object used to send the queries
- @extra => the sql fragment added to the create table sentences
- @asis => the expected result when the field is copied as is

2.20.13 PDO MySQL not null

```
public function test_pdo_mysql_not_null(): void
```

This function checks that the pdo_mysql driver can copy the rows of a table with nulls to a table with a not null default 0 field, this engine converts the nulls to the default value when the field is copied as is.

2.20.14 PDO SQLite not null

```
public function test_pdo_sqlite_not_null(): void
```

This function checks that the pdo_sqlite driver can copy the rows of a table with nulls to a table with a not null default 0 field, this engine fails when the field is copied as is.

2.20.15 Database driver

```
public function test_database(): void
```

This function checks the correctness of the database driver by creating a database connection, sendint queries validating the expected results and closing the connection.

2.21 Test datetime

```
test_datetime.php
```

This test performs some tests to validate the correctness of the datetime functions

2.21.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.21.2 Main class of this unit test

```
final class test_datetime extends TestCase
```

2.21.3 datetime test

```
public function test_datetime(): void
```

This test performs some tests to validate the correctness of the datetime functions

2.22 Test dbschema

```
test_dbschema.php
```

This test performs some tests to validate the correctness of the dbschema functions

2.22.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.22.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.22.3 Main class of this unit test

```
final class test_dbschema extends TestCase
```

2.22.4 dbschema test

```
public function test_dbschema(): void
```

This test performs some tests to validate the correctness of the dbschema functions

2.22.5 sql test

```
public function test_sql(): void
```

This test performs some tests to validate the correctness of the sql functions

2.23 Test dbstats

```
test_dbstats.php
```

This test performs some tests to validate the correctness of the dbstats functions, it covers both the recording side (php/lib/dbstats.php, loaded on demand by db_query() when debug/slowquerystats is enabled) and the app side that exposes the recorded rows (apps/common/php/dbstats.php), since both files work over the same isolated tbl_slowqueries sqlite table

2.23.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.23.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.23.3 Main class of this unit test

```
final class test_dbstats extends TestCase
```

2.23.4 dbstats test

```
public function test_dbstats(): void
```

This test performs some tests to validate the correctness of the dbstats functions

2.24 Test emails

```
test_emails.php
```

This test performs some tests to validate the correctness of the emails functions

2.24.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.24.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.24.3 Main class of this unit test

```
final class test_emails extends TestCase
```

2.24.4 IMAP cleanup

```
private function imap_cleanup(): void
```

This function removes the IMAP accounts left by a previous execution of the IMAP tests that not finished, with its emails and its operations, to not disturb the other tests; it is called at the beginning of the first test and of the IMAP tests

2.24.5 Mock server start

```
private function mock_start(array $script): array
```

This function starts the scripted mock server (utest/php/scripted.php) as a child process, used to reproduce the responses that the real servers never send, and returns an array with the process, the pipes and the port used by the server

- @script => the script of the mock server

2.24.6 Mock server stop

```
private function mock_stop(array $server): void
```

This function waits the end of the mock server and releases its resources

- @server => the array returned by mock_start

2.24.7 emails test

```
public function test_emails(): void
```

This test performs some tests to validate the correctness of the emails functions

2.24.8 folders test

```
public function test_folders(): void
```

This test performs some tests to validate the correctness of the emails folders functions

2.24.9 socket test

```
public function test_socket(): void
```

This test performs some tests to validate the correctness of the socket class used by the mail protocol clients

2.24.10 pop3 test

```
public function test_pop3(): void
```

This test performs some tests to validate the correctness of the pop3 class using the GreenMail server (make teststart)

2.24.11 imap test

```
public function test_imap(): void
```

This test performs some tests to validate the correctness of the imap class using the GreenMail and the Dovecot servers (make teststart) and the scripted mock server

2.24.12 accounts test

```
public function test_accounts(): void
```

This test performs some tests to validate the correctness of the test of the connection of the accounts using the GreenMail and the Dovecot servers (make teststart)

2.24.13 getmail imap test

```
public function test_getmail_imap(): void
```

This test performs some tests to validate the correctness of the synchronization of the IMAP accounts using the Dovecot server (make teststart) with a new user, that starts with an empty mailbox

2.25 Test error

```
test_error.php
```

This test performs some tests to validate the correctness of the error functions

2.25.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.25.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.25.3 Main class of this unit test

```
final class test_error extends TestCase
```

2.25.4 error function test

```
public function test_error(): void
```

This test performs some tests to validate the correctness of the error functions

2.26 Test exec

```
test_exec.php
```

This test performs some tests to validate the correctness of the exec functions

2.26.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.26.2 Main class of this unit test

```
final class test_exec extends TestCase
```

2.26.3 exec test

```
public function test_exec(): void
```

This test performs some tests to validate the correctness of the exec functions

2.27 Test export

```
test_export.php
```

This test performs some tests to validate the correctness of the export functions

2.27.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.27.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.27.3 Main class of this unit test

```
final class test_export extends TestCase
```

2.27.4 Get data

```
private function get_data($nohead, $length): array
```

This function tries to create an array with rows using the numbers.csv file as source, it is able to add or remove the head row and to select the size of the result array of rows

- @nohead => the nohead field used in the import_file function
- @length => the size of the rows array

2.27.5 export xml test

```
public function test_export_xml(): void
```

This test performs some tests to validate the correctness of the export functions

2.27.6 export csv test

```
public function test_export_csv(): void
```

This test performs some tests to validate the correctness of the export functions

2.27.7 export xlsx test

```
public function test_export_xlsx(): void
```

This test performs some tests to validate the correctness of the export functions

2.27.8 export xls test

```
public function test_export_xls(): void
```

This test performs some tests to validate the correctness of the export functions

2.27.9 export ods test

```
public function test_export_ods(): void
```

This test performs some tests to validate the correctness of the export functions

2.27.10 //~ * export bytes test

```
//~ public function test_export_bytes(): void
```

//~ * This test performs some tests to validate the correctness //~ * of the export functions

2.27.11 export edi test

```
public function test_export_edi(): void
```

This test performs some tests to validate the correctness of the export functions

2.27.12 export json test

```
public function test_export_json(): void
```

This test performs some tests to validate the correctness of the export functions

2.27.13 tree2array test

```
public function test_export_tree2array(): void
```

This test performs some tests to validate the correctness of the tree2array function

2.27.14 getkeys test

```
public function test_export_getkeys(): void
```

This test performs some tests to validate the correctness of the getkeys function

2.27.15 external exec

```
public function test_export_external(): void
```

This test performs some tests to validate the correctness of the external exec

2.28 Test file

```
test_file.php
```

This test performs some tests to validate the correctness of the file functions

2.28.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.28.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.28.3 Main class of this unit test

```
final class test_file extends TestCase
```

2.28.4 file test

```
public function test_file(): void
```

This test performs some tests to validate the correctness of the file functions

2.29 Test files

```
test_files.php
```

This test performs some tests to validate the correctness of the files feature

2.29.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.29.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.29.3 Main class of this unit test

```
final class test_files extends TestCase
```

2.29.4 files test

```
public function test_files(): void
```

This test performs some tests to validate the correctness of the files feature

2.29.5 notes test

```
public function test_notes(): void
```

This test performs some tests to validate the correctness of the notes feature

2.29.6 fileslog test

```
public function test_fileslog(): void
```

This test performs some tests to validate the correctness of the fileslog feature

2.30 Test gc

```
test_gc.php
```

This test performs some tests to validate the correctness of the gc functions

2.30.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.30.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.30.3 Main class of this unit test

```
final class test_gc extends TestCase
```

2.30.4 gc exec

```
public function test_gc_exec(): void
```

This test performs some tests to validate the correctness of the gc functions

2.30.5 gc upload

```
public function test_gc_upload(): void
```

This test performs some tests to validate the correctness of the gc functions

2.30.6 gc upload

```
public function test_gc_trash(): void
```

This test performs some tests to validate the correctness of the gc functions

2.31 Test gdlib

```
test_gdlib.php
```

This test performs some tests to validate the correctness of the gdlib feature

2.31.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.31.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.31.3 Main class of this unit test

```
final class test_gdlib extends TestCase
```

2.31.4 gdlib test

```
public function test_gdlib(): void
```

This test performs some tests to validate the correctness of the gdlib feature

2.32 Test getdata

```
test_getdata.php
```

This test performs some tests to validate the correctness of the getdata functions

2.32.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.32.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.32.3 Main class of this unit test

```
final class test_getdata extends TestCase
```

2.32.4 getdata test

```
public function test_getdata(): void
```

This test performs some tests to validate the correctness of the getdata functions

2.33 Test gettext

```
test_gettext.php
```

This test performs some tests to validate the correctness of the gettext functions

2.33.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.33.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.33.3 Main class of this unit test

```
final class test_gettext extends TestCase
```

2.33.4 gettext test

```
public function test_gettext(): void
```

This test performs some tests to validate the correctness of the gettext functions

2.34 Test help

```
test_help.php
```

This test performs some tests to validate the correctness of the help functions

2.34.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.34.2 Loading helper function

```
require_once 'apps/common/php/help.php';
```

This file contains the needed function used by the unit tests

2.34.3 Main class of this unit test

```
final class test_help extends TestCase
```

2.34.4 help test

```
public function test_help(): void
```

This test performs some tests to validate the correctness of the help functions

2.35 Test html

```
test_html.php
```

This test performs some tests to validate the correctness of the html functions

2.35.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.35.2 Loading helper function

```
require_once 'apps/emails/php/html.php';
```

This file contains the needed function used by the unit tests

2.35.3 Main class of this unit test

```
final class test_html extends TestCase
```

2.35.4 html test

```
public function test_html(): void
```

This test performs some tests to validate the correctness of the html functions

2.36 Test html2text library

```
test_html2text.php
```

2.36.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.36.2 Main class of this unit test

```
final class test_html2text extends TestCase
```

2.36.3 html2text

```
public function test_html2text(): void
```

This function checks the correctness of the html2text method provided by the soundasleep/html2text library.

2.37 Test IDs

```
test_ids.php
```

This test performs some tests to validate the correctness of the check_ids feature

2.37.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.37.2 Main class of this unit test

```
final class test_ids extends TestCase
```

2.37.3 IDs test

```
public function test_ids(): void
```

This test performs some tests to validate the correctness of the check_ids feature

2.38 Test image

```
test_image.php
```

This test performs some tests to validate the correctness of the barcode, qrcode, captcha and score features

2.38.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.38.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.38.3 Main class of this unit test

```
final class test_image extends TestCase
```

2.38.4 barcode test

```
public function test_barcode(): void
```

This test performs some tests to validate the correctness of the barcode feature

2.38.5 qrcode test

```
public function test_qrcode(): void
```

This test performs some tests to validate the correctness of the qrcode feature

2.38.6 captcha test

```
public function test_captcha(): void
```

This test performs some tests to validate the correctness of the captcha feature

2.38.7 score test

```
public function test_score(): void
```

This test performs some tests to validate the correctness of the score feature

2.39 Test import

```
test_import.php
```

This test performs some tests to validate the correctness of the import functions

2.39.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.39.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.39.3 Main class of this unit test

```
final class test_import extends TestCase
```

2.39.4 import xml test

```
public function test_import_xml(): void
```

This test performs some tests to validate the correctness of the import functions

2.39.5 import csv test

```
public function test_import_csv(): void
```

This test performs some tests to validate the correctness of the import functions

2.39.6 import xlsx test

```
public function test_import_xlsx(): void
```

This test performs some tests to validate the correctness of the import functions

2.39.7 import xls test

```
public function test_import_xls(): void
```

This test performs some tests to validate the correctness of the import functions

2.39.8 import ods test

```
public function test_import_ods(): void
```

This test performs some tests to validate the correctness of the import functions

2.39.9 import bytes test

```
public function test_import_bytes(): void
```

This test performs some tests to validate the correctness of the import functions

2.39.10 import edi test

```
public function test_import_edi(): void
```

This test performs some tests to validate the correctness of the import functions

2.39.11 import json test

```
public function test_import_json(): void
```

This test performs some tests to validate the correctness of the import functions

2.39.12 import helper test

```
public function test_import_helper(): void
```

This test performs some tests to validate the correctness of the import functions

2.39.13 external exec

```
public function test_import_external(): void
```

This test performs some tests to validate the correctness of the external exec

2.40 Test indexing

```
test_indexing.php
```

This test performs some tests to validate the correctness of the indexing functions

2.40.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.40.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.40.3 Main class of this unit test

```
final class test_indexing extends TestCase
```

2.40.4 indexing test

```
public function test_indexing(): void
```

This test performs some tests to validate the correctness of the indexing functions

2.41 Test iniset

```
test_iniset.php
```

This test performs some tests to validate the correctness of the iniset functions

2.41.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.41.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.41.3 Main class of this unit test

```
final class test_iniset extends TestCase
```

2.41.4 iniset test

```
public function test_iniset(): void
```

This test performs some tests to validate the correctness of the iniset functions

2.42 Test invoices

```
test_invoices.php
```

This test performs some tests to validate the correctness of the invoices functions

2.42.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.42.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.42.3 Main class of this unit test

```
final class test_invoices extends TestCase
```

2.42.4 invoices test

```
public function test_invoices(): void
```

This test performs some tests to validate the correctness of the invoices functions

2.43 Test log

```
test_log.php
```

This test performs some tests to validate the correctness of the log functions

2.43.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.43.2 Main class of this unit test

```
final class test_log extends TestCase
```

2.43.3 log test

```
public function test_log(): void
```

This test performs some tests to validate the correctness of the log functions

2.44 Test mailparse

```
test_mailparse.php
```

This test performs some tests to validate the correctness of the apps/emails/php/mailparse.php polyfill against the real mailparse PECL extension. It does NOT go through mime_parser_class: it calls the mailparse_* functions (native, since this machine has the real extension) and the polyfill's own __mailparse_*_helper functions side by side, on the same sample emails, and asserts each pair returns the same thing. That keeps this test independent from mime_parser_class.php's own transformation logic, which is free to change without this test needing to track it.

Calling the *_helper functions directly (instead of the plain mailparse_* names) is what lets this run on a machine that has the native extension installed: mailparse.php only defines the plain mailparse_* names when the extension is missing, so on a machine that does have it, those names are already the native functions and can't be used to reach the polyfill's own code. See the header comment of apps/emails/php/mailparse.php for details.

2.44.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.44.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.44.3 Main class of this unit test

```
final class test_mailparse extends TestCase
```

2.44.4 Mailparse test

```
public function test_mailparse(): void
```

This function performs some tests to validate the correctness of the apps/emails/php/mailparse.php polyfill by comparing its output against the real mailparse extension, function by function, over a batch of sample emails

2.44.5 Mailparse edge cases test

```
public function test_mailparse_edge_cases(): void
```

Covers branches that the real-inbox sample in test_mailparse() never happens to exercise: a message/rfc822 part (a forwarded email carried as an attachment), a header repeated more than once, a non-ASCII display/filename value, a null part, and the legacy "addr@host (Display Name)" address form

2.44.6 Scoped part data

```
private function scoped_part_data(array $data): array
```

Reduces a mailparse_msg_get_part_data()/__mailparse_msg_get_part_data_helper() result to the keys mime_parser_class.php actually reads, defaulting the optional ones to '' so an omitted key (native, when absent) and an empty-string key (polyfill, always present) compare equal, matching how mime_parser_class.php treats them via !empty() either way

- @data => raw result from either side

2.44.7 Scoped address

```
private function scoped_address(array $entry): array
```

Reduces a mailparse_rfc822_parse_addresses()/ __mailparse_rfc822_parse_addresses_helper() entry to what mime_parser_class.php actually derives from it. Two differences are normalized away here rather than chased in the polyfill:

- 'is_group': never read by mime_parser_class.php, so it's just dropped from both sides.
- 'display' when there is no real display name: the native extension echoes the address itself as 'display' in that case, while the underlying library (and this polyfill) leaves it as ''. mime_parser_class.php already collapses both to '' via its own 'strcasecmp($disp, $email) !== 0' check, so that's exactly what's replicated here.
- @entry => one entry as returned by either side

2.45 Test math

```
test_math.php
```

This test performs some tests to validate the correctness of the math functions

2.45.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.45.2 Loading helper function

```
require_once 'php/lib/math.php';
```

This file contains the needed function used by the unit tests

2.45.3 Main class of this unit test

```
final class test_math extends TestCase
```

2.45.4 math test

```
public function test_math(): void
```

This test performs some tests to validate the correctness of the math functions

2.46 Test memory

```
test_memory.php
```

This test performs some tests to validate the correctness of the memory functions

2.46.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.46.2 Main class of this unit test

```
final class test_memory extends TestCase
```

2.46.3 memory test

```
public function test_memory(): void
```

This test performs some tests to validate the correctness of the memory functions

2.47 Test mime

```
test_mime.php
```

This test performs some tests to validate the correctness of the mime functions

2.47.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.47.2 Loading helper function

```
require_once 'apps/emails/php/html.php';
```

This file contains the needed function used by the unit tests

2.47.3 Main class of this unit test

```
final class test_mime extends TestCase
```

2.47.4 mime test

```
public function test_mime(): void
```

This test performs some tests to validate the correctness of the mime functions

2.48 Test nssdb

```
test_nssdb.php
```

This test performs some tests to validate the correctness of the nssdb functions

2.48.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.48.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.48.3 Main class of this unit test

```
final class test_nssdb extends TestCase
```

2.48.4 nssdb test

```
public function test_nssdb(): void
```

This test performs some tests to validate the correctness of the nssdb functions

2.48.5 certs test

```
public function test_certs(): void
```

This test performs some tests to validate the correctness of the certs functions

2.48.6 actions test

```
public function test_actions(): void
```

This test performs some tests to validate the correctness of the certs functions

2.49 Test output

```
test_output.php
```

This test performs some tests to validate the correctness of the output functions

2.49.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.49.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.49.3 Main class of this unit test

```
final class test_output extends TestCase
```

2.49.4 output test

```
public function test_output(): void
```

This test performs some tests to validate the correctness of the output functions

2.49.5 colorize test

```
public function test_colorize(): void
```

This test performs some tests to validate the correctness of the colorize functions

2.50 Test password

```
test_password.php
```

This test performs some tests to validate the correctness of the password functions

2.50.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.50.2 Loading helper function

```
require_once 'php/lib/password.php';
```

This file contains the needed function used by the unit tests

2.50.3 Main class of this unit test

```
final class test_password extends TestCase
```

2.50.4 password test

```
public function test_password(): void
```

This test performs some tests to validate the correctness of the password functions

2.51 Test pdf

```
test_pdf.php
```

This test performs some tests to validate the correctness of the pdf functions

2.51.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.51.2 Loading helper function

```
require_once 'php/lib/pdf.php';
```

This file contains the needed function used by the unit tests

2.51.3 Main class of this unit test

```
final class test_pdf extends TestCase
```

2.51.4 pdf test

```
public function test_pdf(): void
```

This test performs some tests to validate the correctness of the pdf functions

2.52 Test perms

```
test_perms.php
```

This test performs some tests to validate the correctness of the perms functions

2.52.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.52.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.52.3 Main class of this unit test

```
final class test_perms extends TestCase
```

2.52.4 Authtoken

```
public function test_authtoken(): array
```

This function execute the authtoken rest request, and must to get the json with the valid token to continue in the nexts unit tests

2.52.5 perms test

```
public function test_perms(array $json): void
```

This test performs some tests to validate the correctness of the perms functions

2.53 Test ping

```
test_ping.php
```

This test performs some tests to validate the correctness of the ping feature

2.53.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.53.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.53.3 Main class of this unit test

```
final class test_ping extends TestCase
```

2.53.4 ping test

```
public function test_ping(): void
```

This test performs some tests to validate the correctness of the ping feature

2.54 Test push

```
test_push.php
```

This test performs some tests to validate the correctness of the push functions

2.54.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.54.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.54.3 Main class of this unit test

```
final class test_push extends TestCase
```

2.54.4 push

```
public function test_push(): void
```

This test performs some tests to validate the correctness of the push functions

2.55 Test mime

```
test_replace.php
```

This test performs some tests to validate the correctness of the str_replace instead of the preg_replace

2.55.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.55.2 Main class of this unit test

```
final class test_replace extends TestCase
```

2.55.3 replace test

```
public function test_replace(): void
```

This test performs some tests to validate the correctness of the str_replace_assoc instead of the replace function

2.56 Test sales

```
test_sales.php
```

This test performs some tests to validate the correctness of the sales functions. The invoices app functions (unmake_matrix_data and set_proforma_invoice) are exercised through the web interface, since these functions are loaded on demand by the XML router and can't be required directly (their name clashes with other apps that define a function with the same name, such as apps/crm/php/quotes.php or apps/users/php/matrix.php)

2.56.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.56.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.56.3 Main class of this unit test

```
final class test_sales extends TestCase
```

2.56.4 Authtoken

```
public function test_authtoken(): array
```

This function execute the authtoken rest request, and must to get the json with the valid token to continue in the nexts unit tests

2.56.5 invoices test

```
public function test_invoices(array $json): void
```

This test performs some tests to validate the correctness of the unmake_matrix_data and set_proforma_invoice functions, exercised through the insert and update rest actions of the invoices app

2.56.6 echarts test

```
public function test_echarts(): void
```

This test performs some tests to validate the correctness of the echarts functions

2.56.7 invoices matrix edge cases test

```
public function test_invoices_matrix_edge_cases(array $json): void
```

This test performs some tests to validate the correctness of the unmake_matrix_data function when it receives edge case inputs: an empty payload, a payload missing the taxes/totals keys, a fully blank new row (must be dropped silently), a blank existing row (must become a deletion marker), a leftover row beyond the submitted matrix (must also become a deletion marker) and a new row that is only partially blank (per-field unset branches)

2.56.8 invoices withholding test

```
public function test_invoices_withholding(array $json): void
```

This test performs some tests to validate the correctness of the unmake_matrix_data function when the lines are using taxes and withholdings, the withholdings must be stored in the lines, in the taxes breakdown using the withholding type and in the totals

2.56.9 taxes test

```
public function test_taxes(): void
```

This test performs some tests to validate the correctness of the taxes functions, used by the invoices and the quotes to manage the taxes and the withholdings of their lines depending on the modes defined by the company

2.56.10 invoices numbering test

```
public function test_invoices_numbering(array $json): void
```

This test performs some tests to validate the correctness of the numbering of the invoices, the next code must be computed using the numeric value of the codes of the current year, and not their text

2.57 Test semaphore

```
test_semaphore.php
```

This test performs some tests to validate the correctness of the semaphore functions

2.57.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.57.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.57.3 Main class of this unit test

```
final class test_semaphore extends TestCase
```

2.57.4 semaphore test

```
public function test_semaphore(): void
```

This test performs some tests to validate the correctness of the semaphore functions

2.58 Test server

```
test_server.php
```

This test performs some tests to validate the correctness of the server functions

2.58.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.58.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.58.3 Main class of this unit test

```
final class test_server extends TestCase
```

2.58.4 server test

```
public function test_server(): void
```

This test performs some tests to validate the correctness of the server functions

2.59 Test sql

```
test_sql.php
```

This test performs some tests to validate the correctness of the sql functions

2.59.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.59.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.59.3 Main class of this unit test

```
final class test_sql extends TestCase
```

2.59.4 SQL test

```
public function test_sql(): void
```

This function performs some tests to validate the correctness of the sql functions

2.60 Test strings

```
test_strings.php
```

This test performs some tests to validate the correctness of the strings functions

2.60.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.60.2 Main class of this unit test

```
final class test_strings extends TestCase
```

2.60.3 strings test

```
public function test_strings(): void
```

This test performs some tests to validate the correctness of the strings functions

2.61 Test mime

```
test_strtr.php
```

This test performs some tests to validate the correctness of the str_replace_assoc instead of the strtr function

2.61.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.61.2 Main class of this unit test

```
final class test_strtr extends TestCase
```

2.61.3 strtr test

```
public function test_strtr(): void
```

This test performs some tests to validate the correctness of the str_replace_assoc instead of the strtr function

2.62 Test system

```
test_system.php
```

This test performs some tests to validate the correctness of the system functions

2.62.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.62.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.62.3 Main class of this unit test

```
final class test_system extends TestCase
```

2.62.4 system test

```
public function test_system(): void
```

This test performs some tests to validate the correctness of the system functions

2.63 Test tokens

```
test_tokens.php
```

This test performs some tests to validate the correctness of the token generator and the token format checker

2.63.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.63.2 Main class of this unit test

```
final class test_tokens extends TestCase
```

2.63.3 Tokens test

```
public function test_tokens(): void
```

This function performs some tests to validate the correctness of the token generator and the token format checker

2.64 Test unoconv

```
test_unoconv.php
```

This test performs some tests to validate the correctness of the unoconv functions

2.64.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.64.2 Loading helper function

```
require_once 'php/lib/unoconv.php';
```

This file contains the needed function used by the unit tests

2.64.3 Main class of this unit test

```
final class test_unoconv extends TestCase
```

2.64.4 Pdf test helper

```
private function test_pdf($input, $size): void
```

This function tries to do the test with unoconv2pdf, checks that the input not exists and the output exists to validate the correctness of the function

- @input => the input file to use in the test
- @size => the size in bytes expected for the output

Notes:

The output is not removed here, it is used as cache by the txt test helper to prevent a second conversion of the same file, and is removed by the caller

The size of the output is checked using a delta because the functions always create the output, that is a void file when the conversion fails

2.64.5 Txt test helper

```
private function test_txt($input, $size): void
```

This function tries to do the test with unoconv2txt, checks that the input not exists and the output exists to validate the correctness of the function

- @input => the input file to use in the test
- @size => the size in bytes expected for the output

2.64.6 unoconv test

```
public function test_unoconv(): void
```

This test performs some tests to validate the correctness of the unoconv functions

2.64.7 ocr test

```
public function test_ocr(): void
```

This test performs some tests to validate the correctness of the ocr functions

2.64.8 commands test

```
public function test_commands(): void
```

This test performs some tests to validate the correctness of the commands functions

2.65 Test upload

```
test_upload.php
```

This test performs some tests to validate the correctness of the upload functions

2.65.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.65.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.65.3 Main class of this unit test

```
final class test_upload extends TestCase
```

2.65.4 Authtoken

```
public function test_authtoken(): array
```

This function execute the authtoken rest request, and must to get the json with the valid token to continue in the nexts unit tests

2.65.5 Addfiles

```
public function test_addfiles(array $json): array
```

This function execute the addfiles rest request, and must to get the json with the correct data

2.65.6 Delfiles

```
public function test_delfiles(array $json): void
```

This function execute the delfiles rest request, and must to get the json with the correct data

2.65.7 Human size

```
public function test_human_size(): void
```

This function only try to execute the get_human_size function

2.66 Test users

```
test_user.php
```

This test performs some tests to validate the correctness of the users functions

2.66.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.66.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.66.3 Main class of this unit test

```
final class test_user extends TestCase
```

2.66.4 Authtoken

```
public function test_authtoken(): array
```

This function execute the authtoken rest request, and must to get the json with the valid token to continue in the nexts unit tests

2.66.5 users test

```
public function test_user(array $json): void
```

This test performs some tests to validate the correctness of the users functions

2.67 Test users

```
test_users.php
```

This test performs some tests to validate the correctness of the users functions

2.67.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.67.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.67.3 Main class of this unit test

```
final class test_users extends TestCase
```

2.67.4 days test

```
public function test_days(): void
```

This test performs some tests to validate the correctness of the days functions

2.67.5 users test

```
public function test_users(): void
```

This test performs some tests to validate the correctness of the users functions

2.67.6 groups test

```
public function test_groups(): void
```

This test performs some tests to validate the correctness of the groups functions

2.68 Test version

```
test_version.php
```

This test performs some tests to validate the correctness of the version related functions

2.68.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.68.2 Main class of this unit test

```
final class test_version extends TestCase
```

2.68.3 version test

```
public function test_version(): void
```

This function performs some tests to validate the correctness of the version related functions

2.69 Test web customers

```
test_web_customers.php
```

This test performs all actions of the customers app suck as: create, insert, list, view, edit, update and delete, using the apache sapi interface

2.69.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.69.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.69.3 Main class of this unit test

```
final class test_web_customers extends TestCase
```

2.69.4 Authtoken

```
public function test_authtoken(): array
```

This function execute the authtoken rest request, and must to get the json with the valid token to continue in the nexts unit tests

2.69.5 Create

```
public function test_create(array $json): array
```

This function execute the creates rest request, and must to get the json with the layout without data

2.69.6 Insert

```
public function test_insert(array $json): array
```

This function execute the insert rest request, to do it send the json with the data that they want to insert and must to get the json with the status and the create_id.

2.69.7 List

```
public function test_list(array $json): array
```

This function execute the list rest request, to do it send the json with the search that they want to use in the list filter and receives the json with the data used to populate the table.

2.69.8 View

```
public function test_view(array $json): array
```

This function execute the view rest request, intended to retrieve the detail of the app with the layout needed to render it.

2.69.9 Edit

```
public function test_edit(array $json): array
```

This function execute the view rest request, intended to retrieve the detail of the app with the layout needed to render it.

2.69.10 Upgrade

```
public function test_update(array $json): array
```

This function execute the update rest request, to do it send the json with the data that they want to update and must to get the json with the status and the updated_id.

2.69.11 Delete

```
public function test_delete(array $json): void
```

This function execute the delete rest request, they must to get the json with the status and the deleted_id.

2.70 Test web invoices

```
test_web_invoices.php
```

This test performs all actions of the invoices app suck as: create, insert, list, view, edit, update and delete, using the apache sapi interface

2.70.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.70.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.70.3 Main class of this unit test

```
final class test_web_invoices extends TestCase
```

2.70.4 Authtoken

```
public function test_authtoken(): array
```

This function execute the authtoken rest request, and must to get the json with the valid token to continue in the nexts unit tests

2.70.5 Create

```
public function test_create(array $json): array
```

This function execute the creates rest request, and must to get the json with the layout without data

2.70.6 Insert

```
public function test_insert(array $json): array
```

This function execute the insert rest request, to do it send the json with the data that they want to insert and must to get the json with the status and the create_id.

2.70.7 List

```
public function test_list(array $json): array
```

This function execute the list rest request, to do it send the json with the search that they want to use in the list filter and receives the json with the data used to populate the table.

2.70.8 View

```
public function test_view(array $json): array
```

This function execute the view rest request, intended to retrieve the detail of the app with the layout needed to render it.

2.70.9 Edit

```
public function test_edit(array $json): array
```

This function execute the view rest request, intended to retrieve the detail of the app with the layout needed to render it.

2.70.10 Upgrade

```
public function test_update(array $json): array
```

This function execute the update rest request, to do it send the json with the data that they want to update and must to get the json with the status and the updated_id.

2.70.11 Delete

```
public function test_delete(array $json): void
```

This function execute the delete rest request, they must to get the json with the status and the deleted_id.

2.71 Test web tokens (first part)

```
test_web_tokens.php
```

This test performs some part of the actions related with the tokens suck as authtoken and checktoken, using the apache sapi interface

2.71.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.71.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.71.3 Main class of this unit test

```
final class test_web_tokens extends TestCase
```

2.71.4 Authtoken

```
public function test_authtoken(): array
```

This function execute the authtoken rest request, and must to get the json with the valid token to continue in the nexts unit tests

2.71.5 Checktoken

```
public function test_checktoken(array $json): array
```

This function execute the checktoken rest request, and must to get the json with the ok about the valid token that you are trying to check

2.71.6 Deauthtoken

```
public function test_deauthtoken(array $json): array
```

This function execute the deauthtoken rest request, and must to get the json with the ok about the valid token that you are deauthenticate

2.71.7 Checktoken ko

```
public function test_checktoken_ko(array $json): void
```

This function execute the checktoken rest request, and must to get the json with the ko about the invalid token that you are trying to check

2.72 Test XML

```
test_xml.php
```

This test performs some tests to validate the correctness of the xml related functions

2.72.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.72.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.72.3 Main class of this unit test

```
final class test_xml extends TestCase
```

2.72.4 XML test

```
public function test_xml(): void
```

This function performs some tests to validate the correctness of the xml related functions

2.73 Test YAML

```
test_yaml.php
```

This test performs some tests to validate the correctness of the api/php/autoload/yaml.php polyfill against the real yaml PECL extension: it calls the yaml_* functions (native, since this machine has the real extension) and the polyfill's own __yaml_*_helper functions side by side, on the same sample files, and asserts each pair returns the same thing.

Calling the *_helper functions directly (instead of the plain yaml_* names) is what lets this run on a machine that has the native extension installed: yaml.php only defines the plain yaml_* names when the extension is missing, so on a machine that does have it, those names are already the native functions and can't be used to reach the polyfill's own code.

2.73.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.73.2 Main class of this unit test

```
final class test_yaml extends TestCase
```

2.73.3 YAML test

```
public function test_yaml(): void
```

This function performs some tests to validate the correctness of the api/php/autoload/yaml.php polyfill by comparing its output against the real yaml extension, function by function, over a batch of sample YAML files

2.74 Test zindex

```
test_zindex.php
```

This test performs some tests to validate the correctness of the zindex functions

2.74.1 Importing namespaces

```
use PHPUnit\Framework\TestCase;
```

2.74.2 Loading helper function

```
require_once 'lib/utestlib.php';
```

This file contains the needed function used by the unit tests

2.74.3 Main class of this unit test

```
final class test_zindex extends TestCase
```

2.74.4 zindex test

```
public function test_zindex(): void
```

This test performs some tests to validate the correctness of the zindex functions